PCCST501 - Computer Networks

Assignment 3: Data-Link Layer, Ethernet and Wireless LANs

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Q1. Data-link layer duties; LLC and MAC

The data-link layer moves frames between two directly connected nodes. It takes packets from the network layer and prepares them for one hop over the physical medium, so that the layers above never deal with raw bits.

- **Framing:** a header and trailer are added so the receiver can tell where a frame starts and ends.
- **Hardware addressing:** source and destination MAC addresses name the two ends of the hop.
- **Error detection:** a checksum (CRC-32 in Ethernet) lets the receiver discard damaged frames.
- **Flow control:** a fast sender is slowed so that a slow receiver is not overrun.
- **Medium access control:** on a shared medium, rules decide who may transmit and when.

To handle these tasks the layer is split into two sub-layers:

	LLC (IEEE 802.2)	MAC (IEEE 802.3 / 802.11)
Sits	On top, next to the network layer	Below LLC, next to the physical layer
Job	Tells which network protocol owns the payload (using SAP/SNAP fields); can offer optional acknowledgement and flow control	Builds and parses frames, applies MAC addresses and FCS, and runs the access method
Depends on medium?	No, same for all LAN types	Yes, differs for wired and wireless
Example	Marks a payload as IPv4 or ARP	CSMA/CD in old Ethernet, CSMA/CA in Wi-Fi

Q2. Ethernet frame format

Figure 1 shows an Ethernet II frame with each field's size in bytes.

Preamble 7	SFD 1	Dest MAC 6	Src MAC 6	Type 2	Payload + pad 46-1500	FCS 4

Figure 1: Ethernet II frame (sizes in bytes)

Field	Meaning
Preamble (7 B)	Seven bytes of alternating 1s and 0s that let the receiver lock onto the sender's clock.
SFD (1 B)	10101011; the last two 1 bits announce that the real frame begins.
Destination MAC (6 B)	Receiver's address: unicast, multicast, or the broadcast address FF:FF:FF:FF:FF:FF.
Source MAC (6 B)	Sender's address; switches read it to learn where a host is.
Type (2 B)	A value of 0x0600 or more is an EtherType (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6). Smaller values give the length in the older 802.3 format.
Payload (46-1500 B)	Upper-layer data. Short data is padded to 46 B. 1500 B is the MTU.
FCS (4 B)	CRC-32 over header and payload; a bad value means the frame is dropped.

The smallest legal frame is 14 + 46 + 4 = 64 bytes and the largest is 14 + 1500 + 4 = 1518 bytes (without a VLAN tag). The preamble and SFD are not counted and are usually absent in captures.

Q3. Wireshark capture and MAC addresses of five frames

Method. On Ubuntu, Wireshark is started and the active interface is selected. Traffic is generated with `ping -c 5 8.8.8.8`, `arp -a` and by opening a web page. The capture is stopped after about 30 seconds and the filter `arp || icmp || dns` is applied. For each frame the Ethernet II section is expanded to read the addresses.

Figure 2 below is a documented example (sample capture) and was not captured by me. It shows what such a capture normally looks like; it is included as a reference as permitted by the course instructor. The MAC and IP addresses are illustrative.

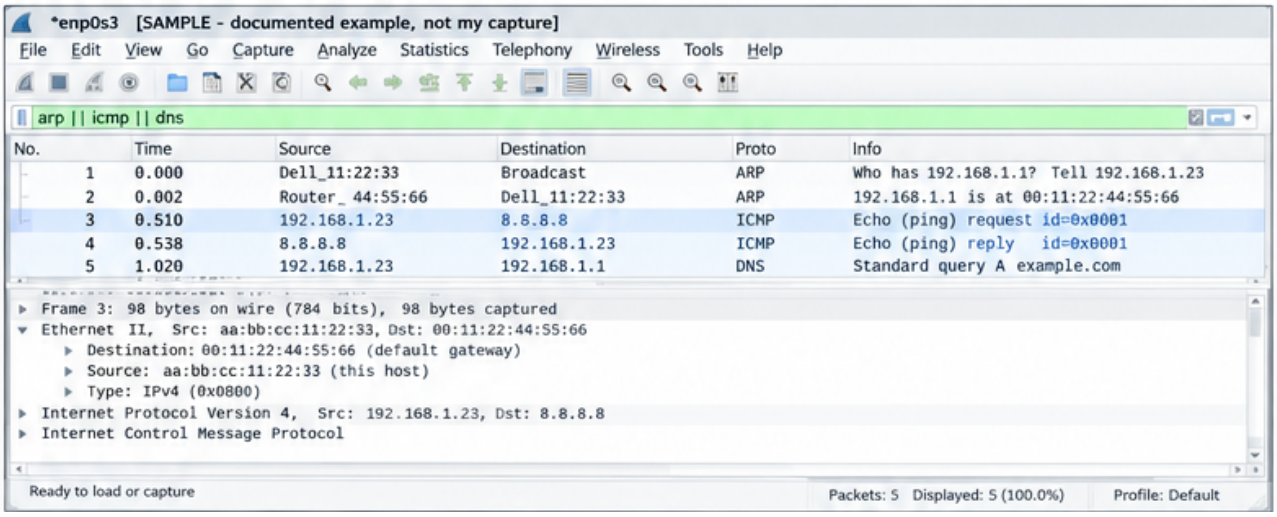


Figure 2: Wireshark sample capture showing five frames and MAC addresses

Frame	Protocol	Source MAC	Destination MAC	EtherType
1	ARP request	aa:bb:cc:11:22:33	ff:ff:ff:ff:ff:ff	0x0806
2	ARP reply	00:11:22:44:55:66	aa:bb:cc:11:22:33	0x0806
3	ICMP request	aa:bb:cc:11:22:33	00:11:22:44:55:66	0x0800
4	ICMP reply	00:11:22:44:55:66	aa:bb:cc:11:22:33	0x0800
5	DNS query	aa:bb:cc:11:22:33	00:11:22:44:55:66	0x0800

Findings. The ARP request is a broadcast because the host does not yet know the gateway's MAC, while the reply is unicast. For the ping to 8.8.8.8 the destination MAC is the gateway's, not Google's, since a MAC address only has meaning on the local link; the IP header keeps the real destination. The first three bytes of each MAC (the OUI) identify the NIC maker.

Q4. How a switch learns MAC addresses

A switch keeps a MAC address table (CAM table) that maps each address to a port. It follows four rules for every incoming frame: **learn** the source address against the arrival port (entries age out, commonly after 300 s); **forward** out of one port if the destination is known; **flood** to all other ports if the destination is unknown or broadcast; **filter** (drop) if the destination is on the arrival port.

Example: four hosts A, B, C, D on ports 1 to 4 of a new switch.

Step	Frame	What the switch does	Table afterwards
1	A to B	Learns A on port 1; B unknown so floods to 2, 3, 4	A:1
2	B to A	Learns B on port 2; A known so sends only to port 1	A:1, B:2
3	C to A	Learns C on port 3; sends only to port 1	A:1, B:2, C:3
4	D to B	Learns D on port 4; sends only to port 2	A:1, B:2, C:3, D:4
5	A to C	All known; sent only to port 3, B and D see nothing	unchanged

On a Cisco switch the table is shown with `show mac address-table`; in Packet Tracer the same behaviour can be watched in simulation mode.

Q5. Hub, bridge and switch compared

Point	Hub	Bridge	Switch
Layer	1 (physical)	2 (data link)	2 (some also 3)
How it forwards	Copies bits to every other port	Uses a MAC table between a few segments, mostly in software	Uses a MAC table per port, in hardware
Collision domains	One for the whole hub	One per segment	One per port
Broadcast domains	One	One	One per VLAN
Duplex	Half only	Half	Full supported
Bandwidth	Shared by all	Shared per segment	Dedicated per port
Today	Obsolete	Replaced by switches	Standard in every LAN

Q6. CSMA/CD and why it matters less now

CSMA/CD was the access method of classic shared-cable (half-duplex) Ethernet. A station first **listens**; if the medium is idle it sends, and keeps **monitoring** while sending. If it notices a collision, it sends a 32-bit **jam signal** so everybody knows, stops, and then applies **binary exponential back-off**: after the n-th collision it waits a random number of slot times between 0 and 2^n - 1 (n limited to 10, giving up after 16 tries). The 64-byte minimum frame (slot time of 512 bit times) guarantees a sender is still transmitting when news of a collision comes back from the far end.

Why it is rarely used now:

- Switches give every host its own point-to-point link, so each port is its own collision domain.
- Full-duplex links have separate send and receive paths, so collisions cannot happen and CSMA/CD is switched off.
- Gigabit and faster Ethernet standards in practice work only in full duplex.
- It survives only with old hubs or a duplex mismatch, which shows up as late collisions and slow transfers.

Q7. CSMA/CA in 802.11 versus CSMA/CD

A radio cannot listen to the channel while it is transmitting, and in the hidden-terminal case two stations cannot hear each other but both reach the access point. Wi-Fi therefore tries to **avoid** collisions with the Distributed Coordination Function:

- Sense the channel (physically, and virtually through the NAV timer).
- If idle, wait a DIFS interval; if busy, wait until it is free.
- Choose a random back-off counter that counts down only while the channel stays idle.
- Send the frame; the receiver answers with an ACK after a short SIFS gap. No ACK means a failure, so the contention window is doubled and the frame is retried.
- Optionally use RTS/CTS so that hidden stations hear the CTS and set their NAV.

Point	CSMA/CD	CSMA/CA
Idea	Detect a collision and recover	Avoid the collision beforehand
Medium	Wired	Wireless
ACK	Not at MAC layer	Required for unicast
Back-off	Only after a collision	Before every send and after failures
Hidden terminal	Not an issue	Handled with RTS/CTS and NAV
Overhead	Low	Higher

Q8. WLAN design for a 12 m x 8 m laboratory

The room holds about 40 workstations plus laptops, roughly 60 devices at once. Three ceiling-mounted access points give enough capacity and coverage.

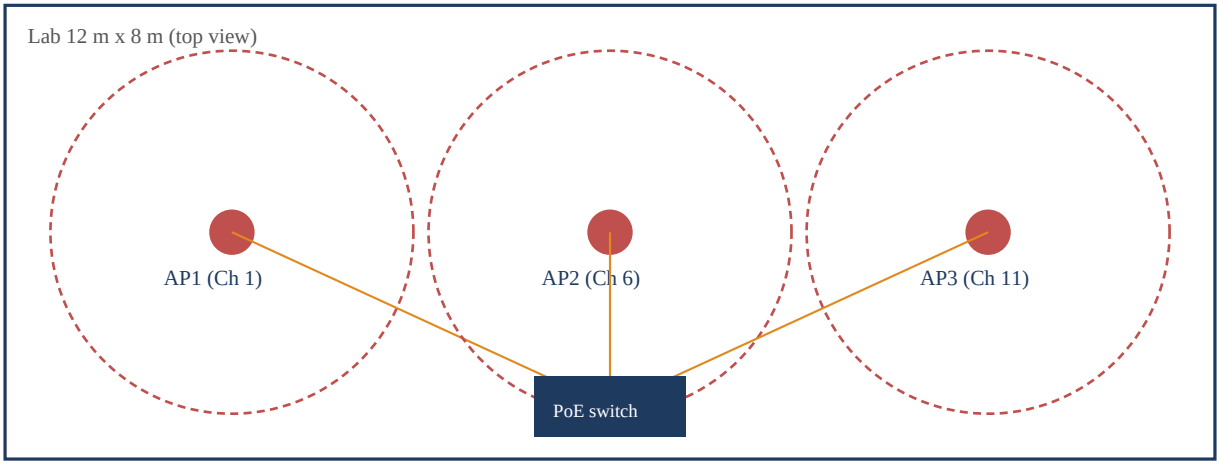


Figure 3: Proposed layout - three APs on channels 1, 6, 11, wired to a PoE switch

Item	Design
AP count and position	3 APs on the ceiling (about 2.7 m), spaced along the room's length with 15-20% cell overlap; away from metal cabinets.
Standard and band	Wi-Fi 5/6 dual band; clients steered to 5 GHz, 2.4 GHz kept for older devices.
Channels	2.4 GHz: 1, 6, 11 on AP1, AP2, AP3. 5 GHz: 20 or 40 MHz channels such as 36, 40, 44.
SSIDs	College-Lab for students/staff and College-Guest (isolated); the same SSID on all APs so devices roam.
Addressing	Lab: 192.168.10.0/24, gateway .1, DHCP .50-.250. APs: static addresses in a management VLAN. Guests: 192.168.20.0/24.
Backbone	APs powered by a Gigabit PoE switch, uplink to a router/firewall doing NAT.

Item	Design
Security	WPA3 (or WPA2-AES), 802.1X with RADIUS where possible, WPS off, default passwords changed, guest client isolation, firmware updates.
Capacity	About 20 clients per AP, comfortably within one radio's limits; guest bandwidth limited.

Q9. Mobile IP and the need for it

An IP address serves both as a name and as a location, because its prefix points to a network. A device that moves to a new network no longer fits the prefix, so traffic still goes to the old network and ongoing TCP sessions break. Mobile IP (RFC 5944 for IPv4) lets the device keep one permanent address while it moves.

Term	Role
Mobile node (MN)	The device that moves.
Home agent (HA)	Router on the home network; tracks the MN and forwards its traffic.
Foreign agent (FA)	Router on the visited network; provides a care-of address.
Home address / care-of address	Permanent address of the MN / temporary address for its current location.

- **Discovery:** agents advertise themselves, so the MN knows it has moved.
- **Registration:** the MN registers its care-of address with the HA.
- **Tunnelling:** the HA catches packets sent to the home address and wraps them in IP-in-IP towards the care-of address, where they are unwrapped.
- **Return path:** the MN replies directly, or through a reverse tunnel when ingress filtering blocks that.

Phones and laptops constantly switch between Wi-Fi and cellular networks, and Mobile IP keeps calls, downloads and remote sessions running. Its drawback is triangle routing, which route optimisation in Mobile IPv6 removes.

Q10. Wired Ethernet frame versus 802.11 frame

Wireshark shows 802.11 frames only when the adapter is in monitor mode, so the standard layouts are compared here (documented layouts, not my own capture).

Preamble 7	SFD 1	Dest 6	Src 6	Type 2	Payload 46-1500	FCS 4

Figure 4a: Ethernet II frame

Frame Ctrl 2	Dur/ID 2	Addr1 6	Addr2 6	Addr3 6	Seq 2	Addr4 6 (opt)	Body 0-2312	FCS 4

Figure 4b: IEEE 802.11 data frame (generic MAC header)

Feature	Ethernet II	802.11
Header	14 bytes	24 bytes (30 with Addr4, 2 more with QoS)
Addresses	2	Up to 4, meaning set by To-DS/From-DS bits
Frame types	Data only at MAC level	Management, control (RTS, CTS, ACK) and data
Control info	EtherType	Frame Control: type/subtype, To/From DS, retry, power management, protected flag
Acknowledgement	None	ACK for every unicast frame
Duration (NAV)	None	Duration field supports virtual carrier sensing
Sequence number	None	Sequence Control for duplicates and fragments

Feature	Ethernet II	802.11
Max payload	1500 B	About 2304-2312 B
Security	None at layer 2 (802.1X / MACsec optional)	WPA2 CCMP / WPA3 GCMP with Protected bit
FCS	CRC-32	CRC-32

Conclusion. Wireless is a shared, error-prone medium, so its frame carries extra addresses, control bits, durations, sequence numbers, acknowledgements and encryption that a switched wired LAN does not need.

References

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- A. S. Tanenbaum and D. Wetherall, *Computer Networks*, Pearson.
- IEEE 802.3 and IEEE 802.11 standards.
- RFC 5944, IP Mobility Support for IPv4, Revised.
- Wireshark User's Guide, https://www.wireshark.org/docs/